

Portfolio Optimisation Methodology

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ABSTRACT

The aim of this project is to help retail investors build an optimal portfolio, customized to their specific risk-aversion parameters.

Keywords: Portfolio Optimisation, Non-linear Model Fitting, Risk Aversion, Utility Mapping

INTRODUCTION

The purpose of this particular document is to explain the methodology and mathematics behind the actual project work. This will serve as an extra-curricular collation of my 2025 Semester Two learnings at university. It is **not** designed to serve as financial advice, nor as a true reflection of an optimal portfolio.

ASSUMPTIONS

The math and means used to derive the utility curves assumes that each investor is rational and risk-averse, roughly adopting the curve:

$$U(W) = -e^{-AW}$$

- Where $U(W)$ refers to Utility (U) as a function of Wealth (W)
- and A represents the coefficient of Absolute Risk Aversion

It is also assumed that each investor should seek to maximise their Mean-Variance Utility:

$$U(\mu, \sigma^2) = \mu - \frac{1}{2}A\sigma^2$$

The data used, a 5-year weekly average of returns, is also assumed to be an accurate measure of all future returns, correlations and variations, for the sake of simplicity.

We must also apply some constraints to our portfolio's construction.

- i. We assume that the sum of the proportions of the portfolio held in each asset is 1.
- ii. We assume that the average retail investor is not buying or selling risk-free assets such as treasury bonds.
- iii. We will adopt two cases
 - a. The investor **can** short assets.
 - b. The investor **can not** short assets.

DATA COLLECTION

Using data collected from Yahoo Finance, we obtain the following matrix of expected returns $\tilde{r}$ and Covariance matrix C :

$$\tilde{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 & 3 & 7 & 8 \\ a & b & c & c & c \\ a & b & c & c & c \\ a & b & c & c & c \\ a & b & c & c & c \end{bmatrix}$$

Where each index tracks to the following ETFs (i.e. Row 1, Column 2 in C corresponds to Covariance between VAS and VAP):

1. VXUS - International Shares (S&P 500 ETF)
2. VAP.AX - Australian Property
3. VHYAX - High Yield
4. VAS.AX - Australian Shares (ASX 200 ETF)
5. GC=F - Gold

PROCESS

Finding Risk Aversion Parameter

As aforementioned, we will first try and fit a Utility curve based on the individuals risk preferences. To do so, we ask the individual a series of questions to determine **Certainty Equivalent Values**. For example:

You have a 50 percent chance of winning \$10,000 and a 50 percent chance of winning nothing. What is the lowest guaranteed amount of cash you would accept instead of taking this gamble?

Say the respondent answered \$3,500, then:

$$U(\text{Certainty Equivalent}) = 0.5 \cdot U(\text{Win}) + 0.5 \cdot U(\text{Lose})$$

Substituting in our utility function:

$$-e^{-A(3500)} = 0.5(-e^{-A(10000)}) + 0.5(-e^{-A(0)})$$

and we solve using Python to find:

$$A \approx 0.00012787$$

We ask the individual a number of these questions, until we have a small dataset of "coordinates" in the $Q_i \sim CE$ plane, such that Q_i is Question i and CE is the corresponding Certainty-Equivalent that the respondent was willing to take. Then, utilising Python and the `curve_fit` function, we start an iterative loop that guesses a value of A . The square of the residual is computed, then the sum of the residuals for each scenario (question) is taken. This loop then repeats with slightly different Risk Aversion parameters A until a minimum SSE is found. This minimum A is what we take as the respondents coefficient of Absolute Risk Aversion.

Case a: Shorting allowed

For our first case, where the individual can short assets, a "pen-and-paper" approach will be adopted as follows:

We will look to find x where

$$x(A) = x_0 + \frac{1}{A}y$$

such that:

$$x_0 = \frac{1}{a}C^{-1}\tilde{e}$$

$$y = C^{-1}\tilde{r} - \frac{b}{a}C^{-1}\tilde{e}$$

and:

- $\tilde{r}$ = as defined previous

- $\tilde{e} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

- $a = \tilde{e}^T C^{-1} \tilde{e}$

- $b = \tilde{r}^T C^{-1} \tilde{e}$

This function is derived through optimisation of the mean-variance utility curve defined previous.

See [here](#) for further information.

Case b: No shorting allowed

In this scenario, we can't easily use the same "pen-and-paper" approach. Rather we use the `PyPortfolioOpt` function to set a "boundary" where $x_i \geq 0$ for $i = 1, 2, 3, 4, 5$. This solver algorithmically maximises the Mean-Variance Utility curve under said condition. It uses a "hill climbing" algorithm known as **Quadratic Programming** that treats $x_i = 0$ as a wall. If the math wants to take this negative, then `PyPortfolioOpt` will slide along the wall to find an optimal solution.

See [here](#) for `PyPortfolioOpt` in depth.

Citations

LaTeX formats citations and references automatically using the bibliography records in your .bib file, which you can edit via the project menu. Use the cite command for an inline citation, like Lees-Miller et al. (2010), and the citep command for a citation in parentheses (Lees-Miller et al., 2010).

Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

Lists

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

...or bullet points ...

- Like this,
- and like this.

...or with words and descriptions ...

Word Definition

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ACKNOWLEDGMENTS

Additional information can be given in the template, such as to not include funder information in the acknowledgments section.

REFERENCES

Lees-Miller, J., Hammersley, J., and Wilson, R. (2010). Theoretical maximum capacity as benchmark for empty vehicle redistribution in personal rapid transit. *Transportation Research Record: Journal of the Transportation Research Board*, (2146):76–83.